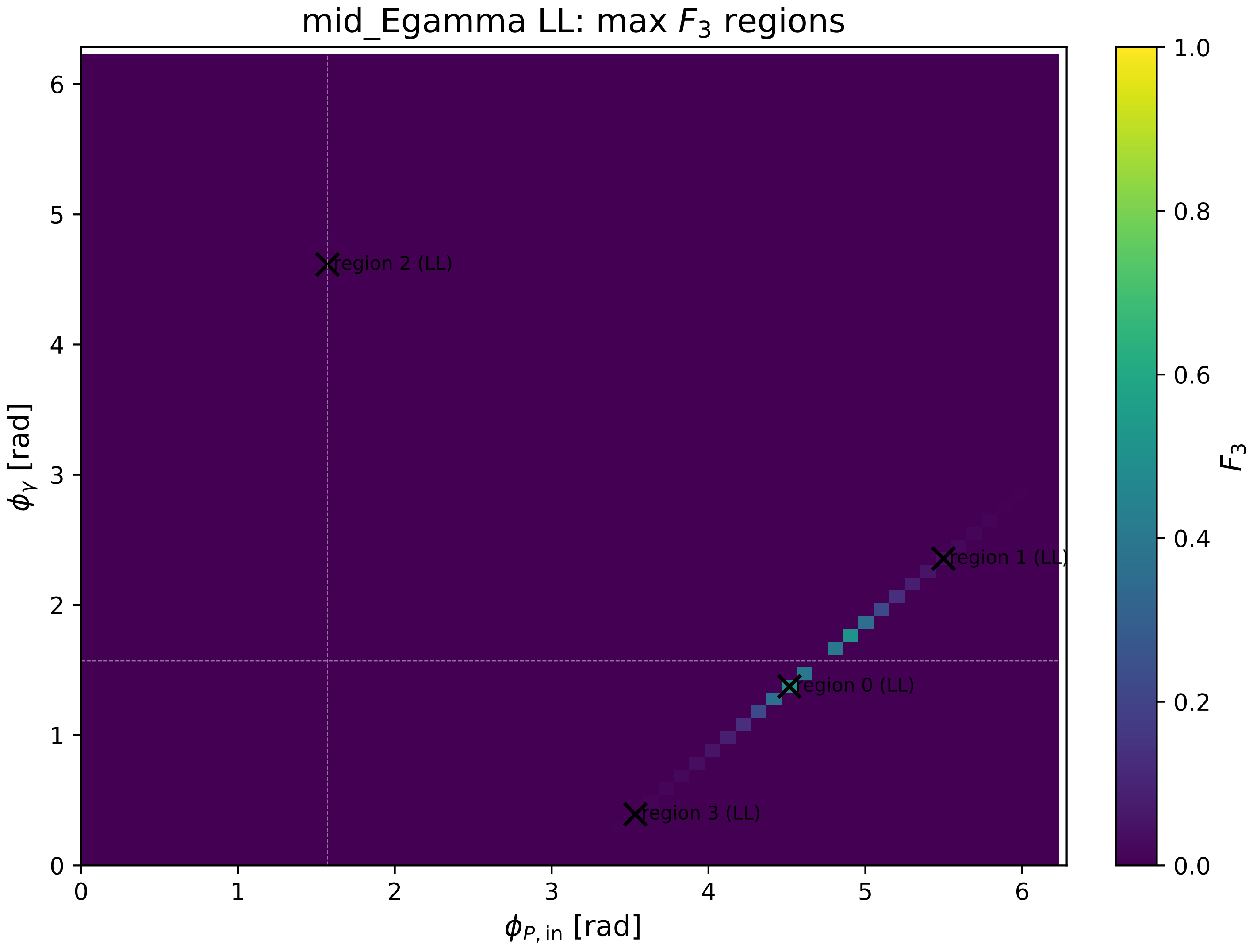
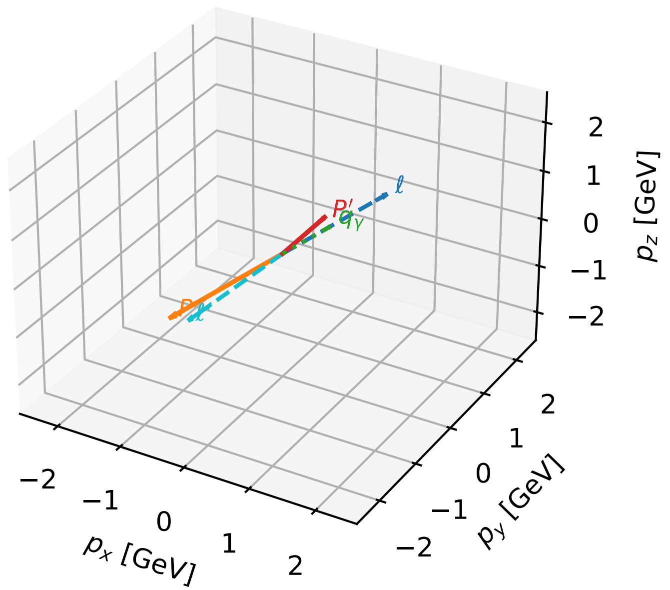

mid_Egamma LL: max F_3 regions
region 2 (LL)
region 1 (LL)
region 0 (LL)
region 3 (LL)
ϕ_γ [rad]
$\phi_{P,\mathrm{in}}$ [rad]
F_3
0
1
2
3
4
5
6
0.0
0.2
0.4
0.6
0.8
1.0

LL characteristic max F_3 configuration

3D momenta

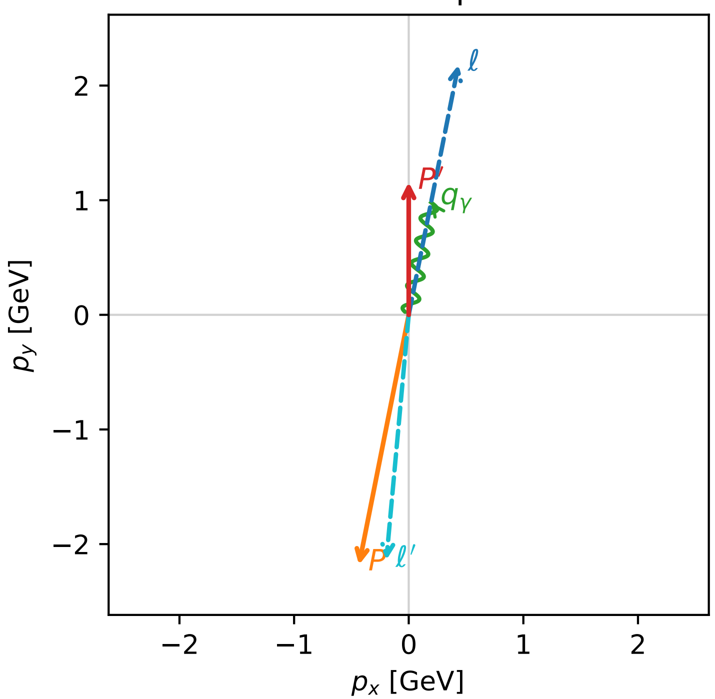



```
f3_mid_egamma_ll_region_0  F_3=0.51066
region: mid_Egamma_LL_region_0
outgoing-state purity: 1
kinematic point: high_s_high_theta_in_mid_Egamma
incoming state: |L_e L_p⟩
```

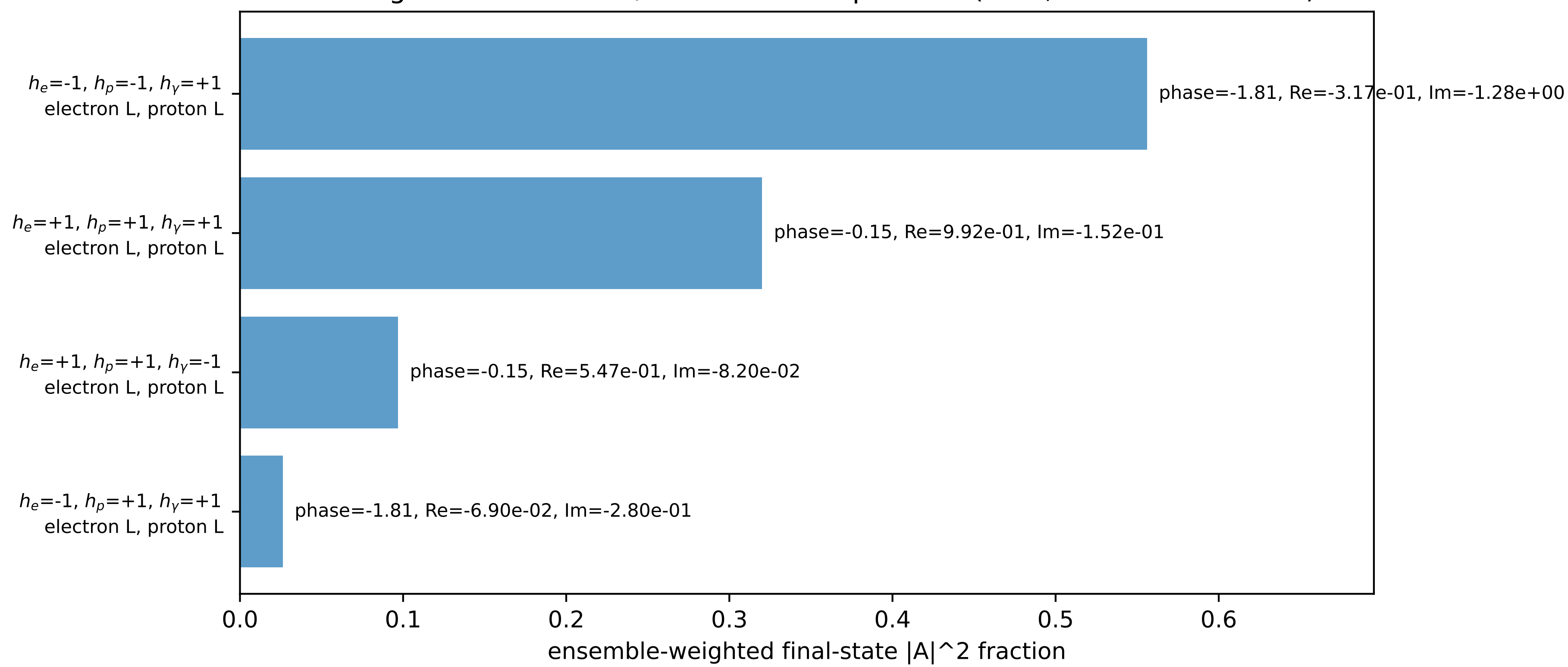
s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22442, $|\vec{P}'|$=1.15835
$\theta_{\rm in}$=1.57079, ϕ_ℓ=1.37445, ϕ_P=4.51604
E_γ=1, ϕ_γ=1.37445
Q^2=19.0593, x_B=0.964142, t=-10.4911, W^2=1.58868, y=0.957942
$\theta(\ell',\gamma)$=173.961 deg

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2244 p_x= 0.4340 p_y= 2.1817 p_z= -0.0000
P E= 2.4141 p_x= -0.4340 p_y= -2.1817 p_z= 0.0000
ℓ' E= 2.1480 p_x= -0.1951 p_y= -2.1391 p_z= 0.0000
P' E= 1.4905 p_x= 0.0000 p_y= 1.1583 p_z= 0.0000
q_γ E= 1.0000 p_x= 0.1951 p_y= 0.9808 p_z= 0.0000

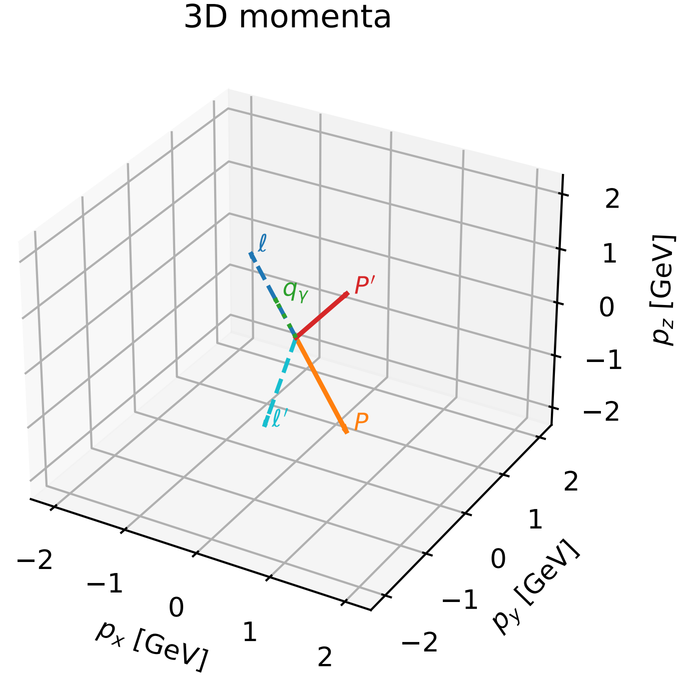
Transverse plane



Leading initial-ensemble/final-state components (N=4, retained=100.0%)



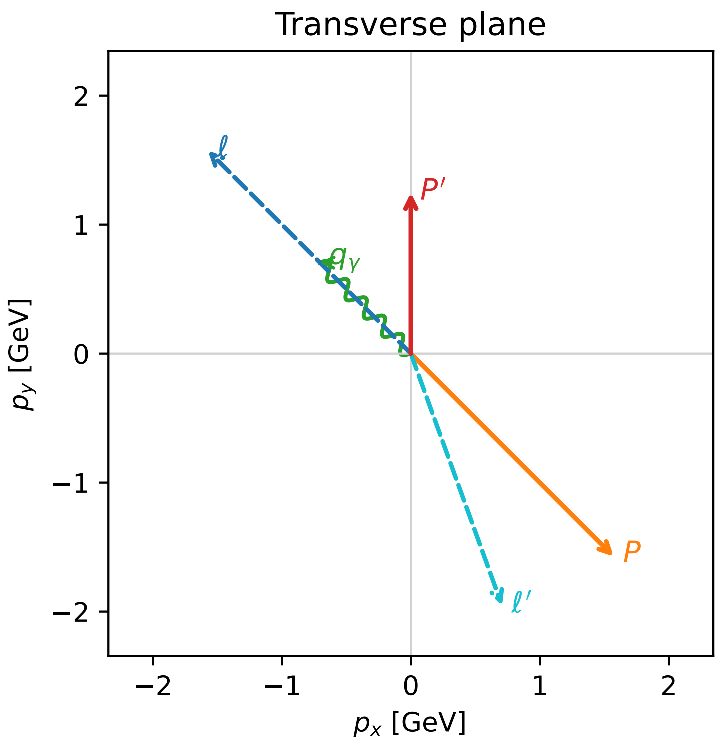
LL characteristic max F_3 configuration

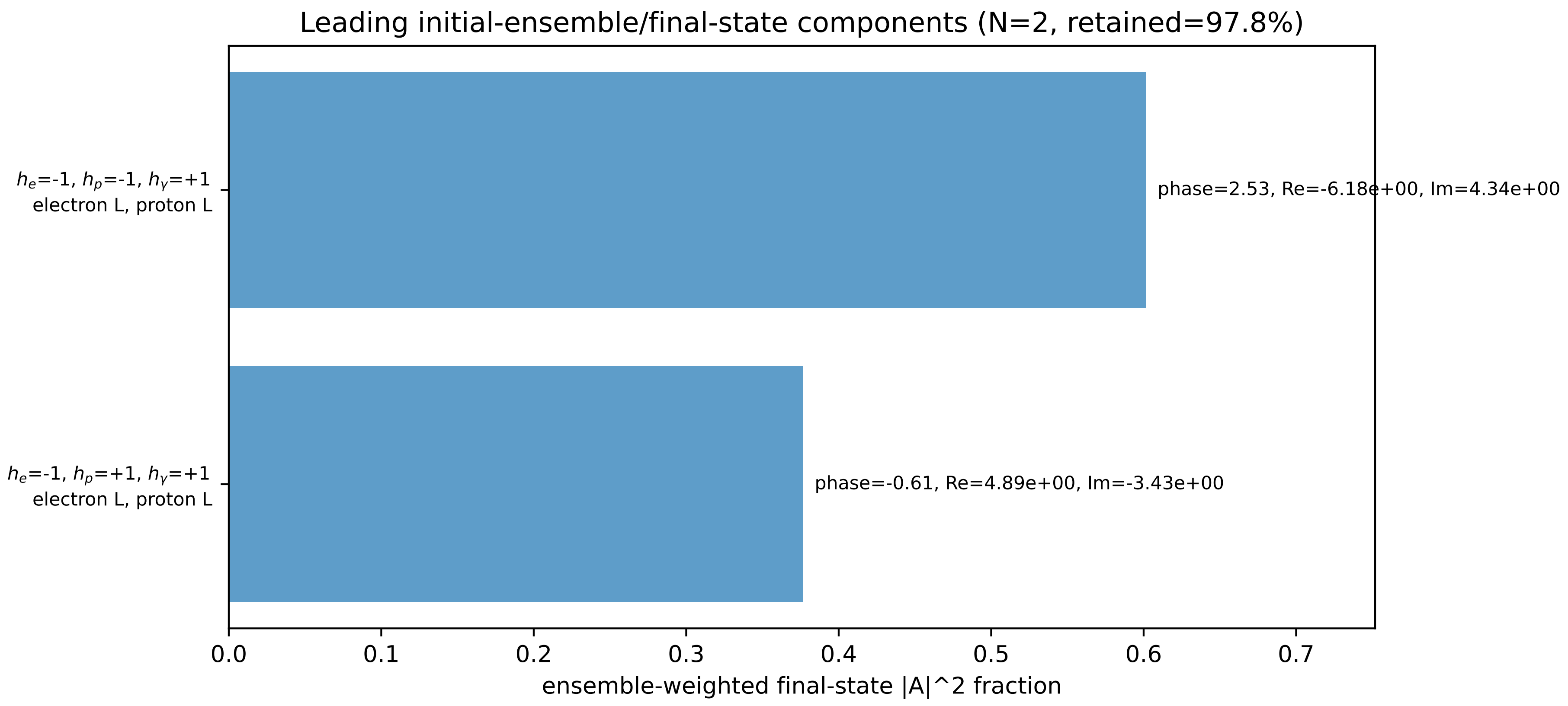


f3_mid_egamma_ll_region_1 F_3=0.0328987
region: mid_Egamma_LL_region_1
outgoing-state purity: 1
kinematic point: high_s_high_theta_in_mid_Egamma
incoming state: $|L_e L_p\rangle$

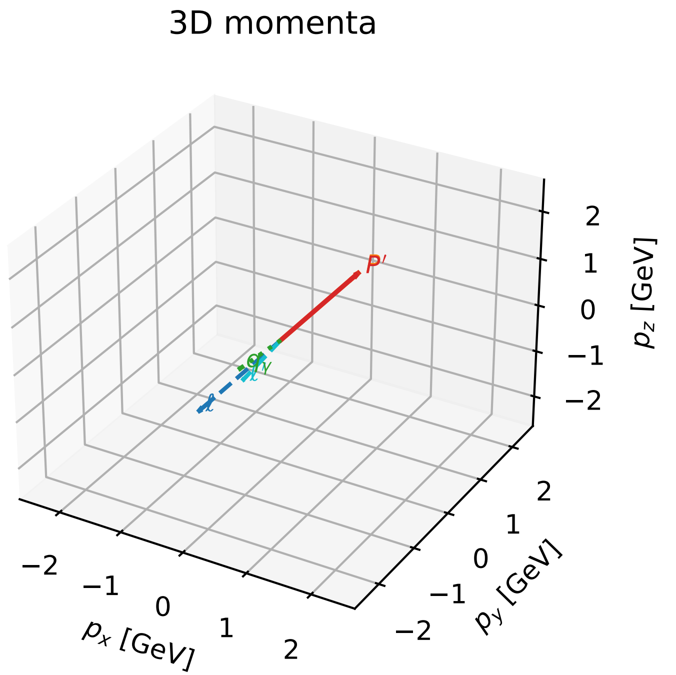
s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22442, $|\vec{P}'|$=1.247
$\theta_{\rm in}$=1.57079, ϕ_ℓ=2.35619, ϕ_P=5.49779
E_γ=1, ϕ_γ=2.35619
Q^2=17.6168, x_B=0.928467, t=-9.69709, W^2=2.23712, y=0.919467
$\theta(\ell', \gamma)$=154.893 deg

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2244 p_x= -1.5729 p_y= 1.5729 p_z= -0.0000
P E= 2.4141 p_x= 1.5729 p_y= -1.5729 p_z= 0.0000
ℓ' E= 2.0781 p_x= 0.7071 p_y= -1.9541 p_z= 0.0000
P' E= 1.5604 p_x= 0.0000 p_y= 1.2470 p_z= 0.0000
q_γ E= 1.0000 p_x= -0.7071 p_y= 0.7071 p_z= 0.0000





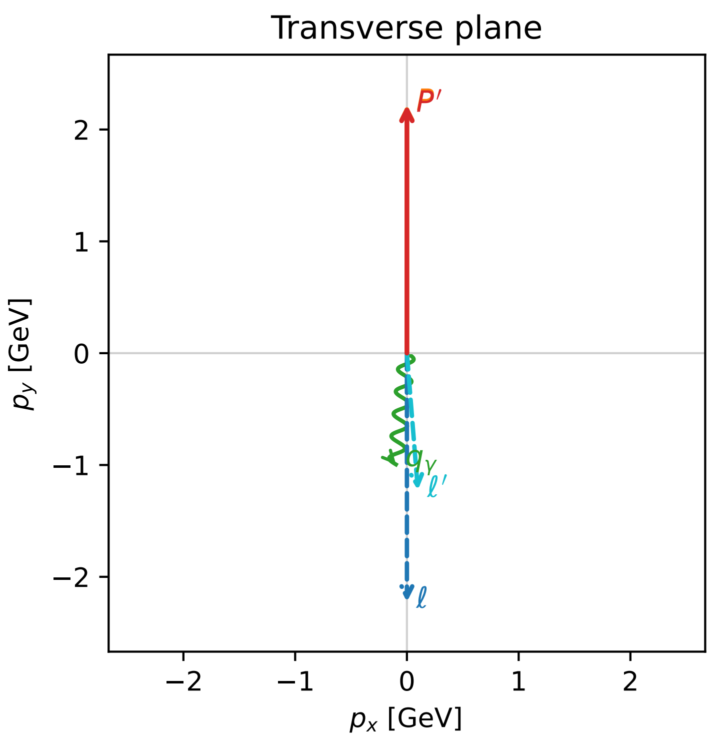
LL characteristic max F_3 configuration

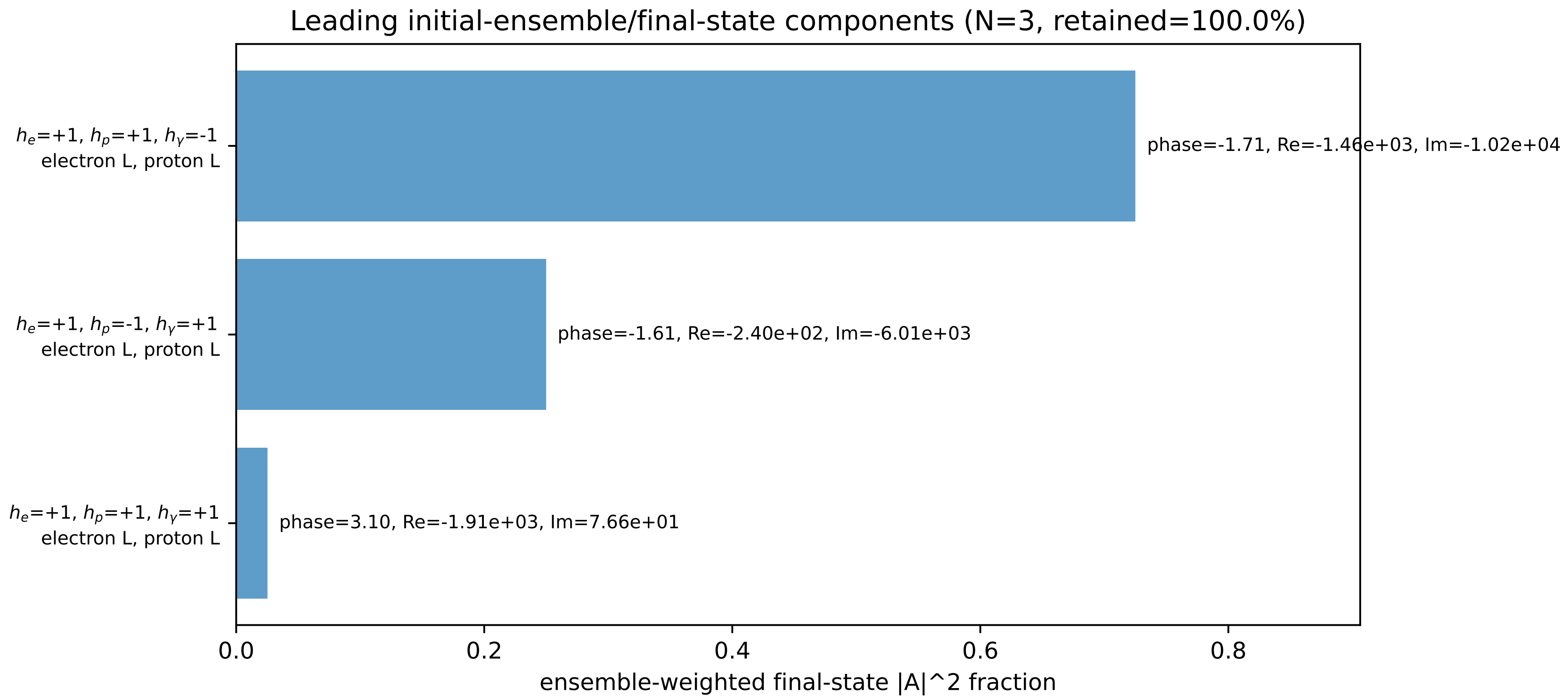



```
f3_mid_egamma_ll_region_2  F_3=0.00773721
region: mid_Egamma_LL_region_2
outgoing-state purity: 1
kinematic point: high_s_high_theta_in_mid_Egamma
incoming state: |L_e L_p⟩
```

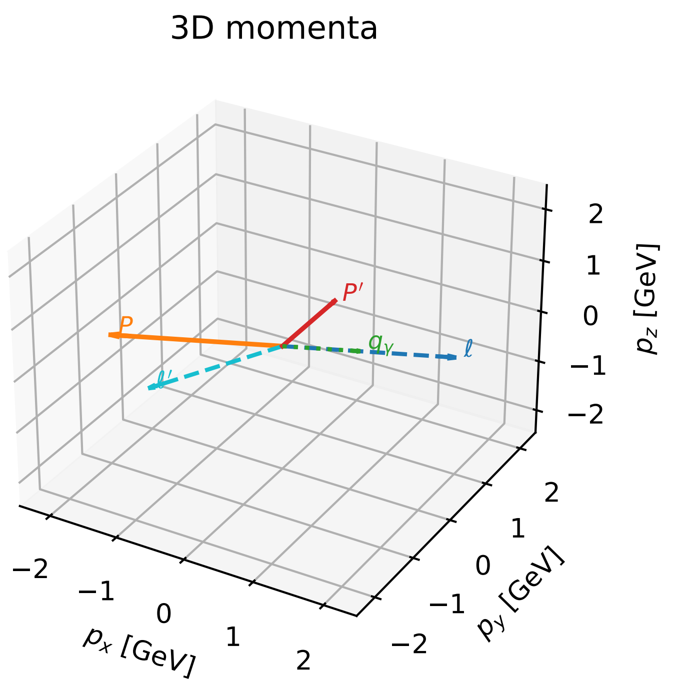
s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22442, $|\vec{P}'|$=2.21987
θ_{in}=1.57079, ϕ_ℓ=4.71239, ϕ_P=1.5708
E_γ=1, ϕ_γ=4.61421
Q^2=0.0174222, x_B=0.00188234, t=-3.12347e-06, W^2=10.118, y=0.448518
$\theta(\ell', \gamma)$=10.2009 deg

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2244 p_x= -0.0000 p_y= -2.2244 p_z= -0.0000
P E= 2.4141 p_x= 0.0000 p_y= 2.2244 p_z= 0.0000
ℓ' E= 1.2286 p_x= 0.0980 p_y= -1.2247 p_z= 0.0000
P' E= 2.4099 p_x= 0.0000 p_y= 2.2199 p_z= 0.0000
q_γ E= 1.0000 p_x= -0.0980 p_y= -0.9952 p_z= 0.0000





LL characteristic max F_3 configuration



f3_mid_egamma_ll_region_3 F_3=0.00662469
region: mid_Egamma_LL_region_3
outgoing-state purity: 1
kinematic point: high_s_high_theta_in_mid_Egamma
incoming state: $|L_e L_p\rangle$

s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22442, $|\vec{P}'|$=1.36819
θ_{in}=1.57079, ϕ_ℓ=0.392699, ϕ_P=3.53429
E_γ=1, ϕ_γ=0.392699
Q^2=15.5854, x_B=0.872842, t=-8.57889, W^2=3.15038, y=0.86528
$\theta(\ell',\gamma)$=140.319 deg

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2244 p_x= 2.0551 p_y= 0.8512 p_z= -0.0000
P E= 2.4141 p_x= -2.0551 p_y= -0.8512 p_z= 0.0000
ℓ' E= 1.9797 p_x= -0.9239 p_y= -1.7509 p_z= 0.0000
P' E= 1.6588 p_x= 0.0000 p_y= 1.3682 p_z= 0.0000
q_γ E= 1.0000 p_x= 0.9239 p_y= 0.3827 p_z= 0.0000